Application of Derivatives

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1 Geometry Formulas

1.1 Surface Area and Volume

Shape	Lateral area	Total area	Volume
Cube	$4a^2$	$6a^2$	a^3
Cuboid	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Sphere		$4\pi r^2$	$\frac{4}{3}\pi r^3$
Cylinder	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone	$\pi r\ell$	$\pi r(r+\ell)$	$\frac{1}{3}\pi r^2 h$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

For a cone, $\ell = \sqrt{r^2+h^2}$.

1.2 Plane Figures

Shape	Area	Perimeter
Circle	πr^2	$2\pi r$
Rectangle	lb	$2(l+b)$
Square	a^2	$4a$
Ring	$\pi(R^2-r^2)$	$2\pi(R+r)$ (both edges)

1.3 Triangles

$$A = \frac{1}{2}bh, \qquad s = \frac{a+b+c}{2}, \qquad A = \sqrt{s(s-a)(s-b)(s-c)}.$$

For an equilateral triangle, $A = \frac{\sqrt{3}}{4}a^2$ and $P = 3a$. In general, $P = a+b+c$.

2 Rate of Change

For $y = f(x)$, the rate of change is dy/dx. Positive derivative indicates increasing behavior and negative derivative indicates decreasing behavior. Units must be retained: for example, if A is in cm^2 and r in cm, dA/dr has units cm.

3 Increasing and Decreasing Functions

For $x_1 < x_2$:

- Increasing: $f(x_1) \le f(x_2)$; strictly increasing: $f(x_1) < f(x_2)$.
- Decreasing: $f(x_1) \ge f(x_2)$; strictly decreasing: $f(x_1) > f(x_2)$.

$$f'(x) > 0 \Rightarrow \text{strictly increasing}, \qquad f'(x) < 0 \Rightarrow \text{strictly decreasing}$$

on an interval where the derivative has the indicated sign.

4 Slope and Critical Points

The slope of the tangent at $x = c$ is $f'(c)$. To locate possible maxima and minima:

1. Compute $f'(x)$.
2. Solve $f'(x) = 0$ and identify points where the derivative does not exist.
3. Divide the interval at these critical points.
4. Determine the sign of $f'(x)$ in each interval.
5. Check the critical points and endpoints for extrema.

4.1 First Derivative Test

Before c	After c	Conclusion
$+$	$-$	Local maximum
$-$	$+$	Local minimum
$+$	$+$	No sign change
$-$	$-$	No sign change

For absolute extrema on a closed interval $[a, b]$, compare function values at all critical points and endpoints.

5 Second Derivative Test

If $f'(c) = 0$:

$$f''(c) < 0 \Rightarrow \text{local maximum at } c,$$
$$f''(c) > 0 \Rightarrow \text{local minimum at } c,$$
$$f''(c) = 0 \Rightarrow \text{test is inconclusive.}$$